POST-INCIDENT MEMORANDUM

To: Billing Operations, Group Finance, IT Service Management
From: C. Eberhart, Batch Operations
Date: 2026-05-21
Subject: INC-5310 — nightly billing batch overran its window on 2026-05-18, delaying 61,400 invoices by 4h 40m

Jira issue	INC-5310
Severity	SEV-3 — delayed, not lost; no customer-facing outage
Environment	Production — batch cluster btc-prod, billing-engine 8.2.1
Occurred	2026-05-18, batch window 22:00–04:00 CEST
Completed	2026-05-19 08:40 CEST (4h 40m past the window)
Author	C. Eberhart · reviewed by the Billing Operations forum on 2026-05-21

1. What happened

The nightly billing batch for 18 May 2026 started on schedule at 22:00 CEST and was expected to finish by 04:00. It finished at 08:40, four hours and forty minutes late. Because invoice dispatch is gated on batch completion, 61,400 invoices that should have been sent before the 06:00 dispatch cut-off went out in the 09:00 run instead. Direct debit files for two partner banks missed their 07:00 submission slot and were submitted the following day.

No invoices were lost, duplicated or mis-stated. The delay was visible to customers only as an invoice arriving later in the day than usual. Group Finance confirmed on 19 May that the month-end close was not affected.

2. Impact

- 61,400 invoices dispatched 4h 40m late (of 208,000 in the run).
- 2 direct debit submissions missed their bank cut-off; CHF 3.1m collected one day later than planned.
- 34 support contacts, all resolved with a standard explanation.
- No financial loss. Interest cost of the one-day collection delay estimated at CHF 210.

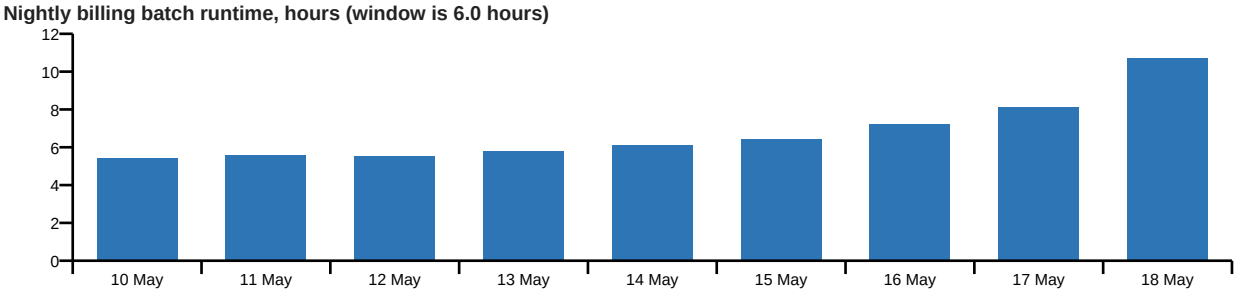

Figure 1 — runtime had been growing for a week before the overrun. The 6-hour window is exceeded from 16 May onward.

3. Timeline (CEST)

When	What happened
2026-05-11	Monthly pricing catalogue loaded. Row count grows from 1.2m to 4.9m following the addition of the partner tariff tables.
2026-05-16 04:38	Batch first exceeds its window, by 38 minutes. The overrun is logged but the alert threshold is set at 120 minutes, so nobody is notified.
2026-05-17 05:06	Second overrun, 66 minutes. Again below the alert threshold.
2026-05-18 22:00	Batch starts on schedule.

When	What happened
2026-05-19 01:12	Step 4 (tariff resolution) enters its long phase; throughput drops to 190 accounts/second from a normal 1,400.
2026-05-19 04:00	Window closes with the batch at 61% complete. Overrun alert fires at the 120-minute mark, 06:00.
2026-05-19 06:04	On-call operator acknowledges; confirms the batch is progressing and elects to let it finish rather than restart.
2026-05-19 06:20	Dispatch team informed that the 06:00 cut-off has been missed. Partner banks notified of the direct debit delay.
2026-05-19 08:40	Batch completes successfully. Invoice dispatch runs at 09:00.
2026-05-21 10:00	Post-incident review held.

4. Root cause

Step 4 of the batch resolves a tariff for every account by scanning the pricing catalogue. The query relies on an index on `(tariff_code, valid_from)`. The partner tariff tables loaded on 11 May quadrupled the catalogue and introduced a large number of rows sharing a single tariff_code, which made that index unselective. The planner switched to a sequential scan per account, turning a linear step into an effectively quadratic one. Runtime therefore grew each night as more partner accounts were activated, crossing the window on 16 May and overrunning badly on 18 May.

The growth was visible in the runtime metric for a week. It was not acted on because the only alert on this job fires at 120 minutes past the window, and the two earlier overruns were 38 and 66 minutes.

5. Action plan

Ref	Action	Owner	Due
OPS-3401	Batch Operations will keep an eye on the nightly runtime and escalate if it happens again.	Batch Operations	Ongoing

The forum discussed indexing changes and an earlier alert threshold, but no owner was available before the quarter-end freeze and no further items were raised at this time.

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2026-05-21